Causal Question

- In one sentence, write the causal question you intend to test. It should be composed of a *response* variable having to do with perception of bubble water and whether it is affected by a given intervention / *explanatory* variable.

Null Hypothesis

- Write down:
  - the corresponding *null hypothesis* that you will gather evidence against
  - the *alternative hypothesis*,
  - and the *significance level*, α, that you intend to use.
- Identify your *test statistic*.

Protocol

Record below the step-by-step protocol that you will use to collect data to bear on the claim above. Be precise. You should be able to hand this off to another group and have them conduct your experiment.

Data

- In the space below, sketch out an empty data frame that stores the data you will have collected per your **Protocol**. The number of rows should reflect your sample size, n, and the number of columns should represent the number of variables that you intend to collect.
- Note also how you are defining your unit of observation.

Exploratory Data Analysis

- Sketch a plot that you will use to use to determine how consistent your data are with your original claim. Please label the axes.
  - Remember: these are plots of data, not null distributions of statistics!
  - Since you do not yet have data, create two sketches of what the plot *might* look like in two scenarios.
- Below each plot, write the ggplot2 code that will create that graphic.

Scenario 1: Data supports H_0 **Scenario 2: Data supports H_A**